

Solutions and Solubility — Paper A

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, your answer must talk about **particles** — solute particles spreading between solvent particles, collisions, or how much the solvent can hold at that temperature.
- Solubility is always given in **g per 100 g of water**. If a question uses a different mass of water, scale it.
- The mark for each question is shown in square brackets on the right.
- Reminder: stirring, crushing and heating change how **fast** something dissolves. Only temperature changes how **much**.

SECTION 1 – THE WORDS

1. For each solution, name the solute and the solvent. [4]

Solution	Solute	Solvent
sea water		
sugar dissolved in tea		
nail varnish dissolved in propanone		
a fizzy drink (carbon dioxide in water)		

2. State what is meant by each word.

(a) soluble [1]

(b) insoluble [1]

(c) saturated solution [1]

- 3.** Sand is stirred into a glass of water; salt is stirred into another. Describe what you would see in each glass after ten minutes, and name the type of mixture in each. [4]

SECTION 2 – PARTICLES

- 4.** Describe, in terms of particles, what happens when sugar dissolves in water. [3]

- 5.** 20 g of salt is dissolved in 80 g of water.

(a) State the mass of the solution. [1]

(b) Explain your answer. [2]

- 6.** A solution of copper sulfate is clear and blue. A student says "clear means nothing is dissolved in it". Explain why the student is wrong. [2]

SECTION 3 – HOW MUCH

- 7.** The solubility of potassium nitrate at 20 °C is 32 g per 100 g of water. A student stirs 40 g into 100 g of water at 20 °C.

(a) State the mass that dissolves and the mass left undissolved. [2]

(b) State whether the solution is saturated, and how you can tell. [2]

8. Give **three** ways to make a solid dissolve faster. [3]

9. State the units of solubility, and explain why a solubility must always be given with a temperature. [4]

End of paper. Check every answer has a unit or a form the question actually asked for.